Tarundeep Arora

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SUMMARY

Data Science undergraduate at Simon Fraser University seeking co-op in software, data, or analytics. Analysed datasets of 70M and 18M records with Python and PySpark, applying statistical testing and machine learning (82% classifier accuracy), and built full stack apps in React, FastAPI, and PostgreSQL, including a deployed app with 100 automated tests and 96%+ coverage.

EDUCATION

Bachelor of Science, Data Science **September 2023 to 2028**
Simon Fraser University, Burnaby, BC
Coursework: Computational Data Science, Data Structures and Algorithms, Databases, Software Engineering, Linear Algebra

PROJECTS

Shrinkflation in Canadian Groceries **SFU CMPT 353**
Python, Pandas, SciPy, scikit-learn, TF-IDF *Course Project*

- Built a pipeline over about 70M scraped price records (3.5GB, 8 chains), fixing a 100x price error in 42,000+ rows.
- Matched products with TF-IDF cosine similarity and a 14 day replacement rule, flagging 2,214 shrinkflation candidates.
- Found a 19.0% median unit price increase for candidates versus 0.3% for controls (Mann-Whitney U, $p < 0.001$).
- Showed the effect differed across chains (Kruskal-Wallis, $p < 0.001$) and validated the work with 78 automated tests.
- Benchmarked a Jevons index against StatCan Food CPI; geometric averaging cut its rise from 13.7% to 1.8% (CPI: 6.5%).

Reddit Corpus Analysis **SFU CMPT 353**
PySpark, Hadoop/HDFS, Pandas, scikit-learn *Course Project*

- Built a PySpark pipeline on SFU's Hadoop cluster, reducing 18M Reddit records (150 subreddits) to 2M observations.
- Showed scores differ by subreddit (Kruskal-Wallis) and weekday comments run higher (Mann-Whitney U), $p < 0.001$.
- Trained a gradient boosted classifier for above median engagement: 82% accuracy, 0.80 F1, and 0.87 AUC.
- Identified subreddit size, posting time, title length, and early engagement as the top predictors of post success.
- Fit a power law model of regional subreddit activity against population (exponent about 0.85, R^2 about 0.78).

Daily Dilemma, Web Application and User Service **October 2025 - December 2025**
React, FastAPI, PostgreSQL, Docker, PyTest *SFU CMPT 276 (A+)*

- Led backend for a daily dilemma game, building 15 FastAPI and PostgreSQL REST endpoints with JWT authentication.
- Built analytics across 5 metrics, including prediction accuracy, participation history, and majority alignment.
- Built React components and the frontend backend integration for 3 game modes: Prediction, Opinion, and Party.
- Refreshed live community results with 5 second polling, switching to real majority results after 35 responses.
- Wrote 100 PyTest tests (96%+ coverage) and deployed with Docker Compose on Render for 50+ users and 500+ votes.

Health First Habit Tracking App **December 2025**
TypeScript, React, Next.js, Prisma, PostgreSQL, Indexed DB *Personal Project*

- Built a local first habit tracker with 10 screens, including daily goals, mood tracking, daily wins, and Bad Day Mode.
- Used IndexedDB for full offline use, with partial completion tracking and dark mode and large text options.
- Built Next.js and Prisma REST routes on PostgreSQL for registration, login, device linking, and cloud backup.
- Secured user accounts with bcryptjs password hashing and JWT authentication stored in HTTP only cookies.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C++, R
Data and ML: Pandas, NumPy, SciPy, scikit-learn, PySpark, Hadoop/HDFS, Hypothesis Testing, Regression, TF-IDF
Web and Backend: React, Next.js, FastAPI, REST APIs, JWT Authentication, PostgreSQL, SQL Alchemy, Prisma
Tools and DevOps: Docker, Docker Compose, Git/GitHub, PyTest, Render, Linux
Spoken Languages: English (Fluent), Hindi (Fluent), Punjabi (Native), Spanish (Intermediate)

WORK EXPERIENCE

Data Analyst, Fortune Law **June 2022 - January 2023**

- Maintained and updated client records and Excel spreadsheets, keeping information organized and accessible for day-to-day operations.
- Reviewed client records for missing or inconsistent information and updated records to maintain data accuracy.